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Inventor(s)	Strahl; Shane et al.

Radial load measuring apparatus

Abstract

A radial load measuring apparatus is disclosed, which may include a rotational member rotatably supported to rotate about an axis of rotation and configured to be subjected to a radial load, and at least one support member interfacing with the rotational member. Each support member may have a contact interface that is oriented at an oblique angle relative to a reference plane comprising the axis of rotation, such that the radial load applied to the rotational member is translated into a proportional axial force applied to the at least one support member in an axial direction. The radial load measuring apparatus may include a force sensor configured to measure the proportional axial force applied to the at least one support member from which to determine a magnitude of the radial load.

Inventors: Strahl; Shane (Jenks, OK), Hendrix; Floyd James (Coweta, OK), Miller; Jonathan (Skiatook, OK), Nelson; Christopher A. (Lake Preston, SD)

Applicant: BPG-ARROWHEAD WINCH INC. (Broken Arrow, OK)

Family ID: 96950428

Assignee: BPG-ARROWHEAD WINCH INC. (Broken Arrow, OK)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4015468	12/1976	Simon	N/A	N/A
5488871	12/1995	Harbottle	73/862.55	G01L 5/12
9476802	12/2015	Sato	N/A	N/A
2023/0221200	12/2022	Blake	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2004003918	12/2003	JP	F16C 19/522
2006214931	12/2005	JP	N/A
2010216654	12/2009	JP	N/A
2010216655	12/2009	JP	N/A
WO-0177634	12/2000	WO	G01L 5/00

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Application No. PCT/US2025/028206; mailed Jul. 1, 2025; 12 pages. cited by applicant

Primary Examiner: Hollington; Octavia

Attorney, Agent or Firm: Seed Intellectual Property Law Group LLP

Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure relates generally to radial load measuring devices, specifically measuring devices for measuring a radial load (e.g., wire load) applied to a rotating drum of a winch or a hoist.

Description of the Related Art

(2) Winches and hoists may be used to lift and/or pull an object, which may subject the winch or hoist to significant loads during operation. Both the torque and radial load that a winch or a hoist can withstand may be an important factor to consider when designing, selecting and/or using a winch or hoist. However, measuring a radial load being applied to a winch or hoist may be difficult to accurately determine and known systems may be overly complex or difficult to implement in cost effective manner. A solution for measuring a radial load being applied to a winch or a hoist throughout operation could provide significant improvements to the design, testing and use of winches and hoists.

BRIEF SUMMARY

(3) Embodiments described herein include radial load measuring apparatuses and related radial load measuring techniques that are well suited to measure radial loads applied to a rotating member, such as a wire load applied to a rotating drum of a winch or a hoist.

(4) For example, a radial load measuring apparatus according to an example embodiment may be summarized as including a rotational member rotatably supported to rotate about an axis of rotation and configured to be subjected to a radial load, at least one support member interfacing with the rotational member, and at least one force sensor. Each support member may have a contact interface that is oriented at an oblique angle relative to a reference plane comprising the axis of rotation such that the radial load applied to the rotational member is translated into proportional axial forces that are applied to the pair of support members in opposite axial directions. The force sensor may be configured to measure at least one of the proportional axial forces applied to the support members from which to determine a magnitude of the radial load.

(5) In some embodiments, the support member may include at least a portion of a taper roller bearing assembly or other bearing assembly. In some embodiments, the support member may include a cone of a taper roller bearing assembly and the rotational member may be fitted with a cup of the taper roller bearing assembly. In such instances, during operation, the cone of the taper roller bearing assembly may be able to translate away from the cup of the taper roller bearing assembly to engage or interact with a force sensor. In this manner, the force sensor may be configured to measure the proportional axial force applied to the support member via displacement of the cone of the taper roller bearing assembly relative to the cup of the taper roller bearing assembly.

(6) In some embodiments, the force sensor may be any of a capacitive load cell, a pneumatic or hydraulic piston sensor, and a strain gauge load cell that interacts with the support member, or any other type of force sensor. The force sensor or sensors may be arranged to measure the proportional axial force that is transferred by the support members with the obliquely aligned contact interface to redirect the radial load applied to the rotational member into the proportional axial force.

(7) In some embodiments, the proportional axial force applied to the support member that is measured by the force sensor may be directly proportional to a magnitude of the radial load applied to the rotational member. In addition, a relationship between the proportional axial force and the magnitude of the radial load may be dictated by an angle or angles of the contact interfaces of the support members and a position of the support members in an axial direction.

(8) In some embodiments, the rotational member may be a drum of a winch or hoist and the radial load applied to the drum may be a wire load. Accordingly, embodiments may provide active, real-time radial wire load measurements during operation or use of a winch or a hoist.

(9) In some embodiments, a pair of support members may be provided and each support member of the pair of support members may interface with a respective one of the opposing ends of the rotational member. In this manner, the rotational member, e.g., drum, may be rotatably supported in a symmetric manner. In other instances, the rotational member, e.g., drum, may be supported asymmetrically and the load calculations contemplated herein may account for the same.

(10) In some embodiments, the oblique angle of the contact interface of each support member may be uniform such that proportional axial forces are divided evenly or substantially evenly in opposing axial directions. The force sensor or sensor(s) may be arranged at only one end of the rotational member or, alternatively, at both of opposing ends of the rotational member. Having sensor(s) at both ends of the rotational member may enable enhanced accuracy of the load measurement. Providing sensor(s) at only one end of the rotational member may lead to a more compact apparatus that is less complex and less costly.

(11) A method of measuring a radial load applied to a rotational member according to an example embodiment may be summarized as including, during application of the radial load, measuring at least one proportional axial force applied to support members and calculating a magnitude of the radial load based at least in part on the measurement of the proportional axial force. The rotational member may be rotatably supported to rotate about an axis of rotation and may be supported by at least one support member interfacing with the rotational member. In some instances, a pair of support members may be provided and each support member may include a contact interface that is

oriented at an oblique angle relative to a reference plane comprising the axis of rotation, such that the radial load applied to the rotational member is translated into proportional axial forces that are applied to the pair of support members in opposite axial directions.

(12) In some embodiments, the calculating of the magnitude of the radial load may be based in part on the oblique angle of the contact interface of the support members. In some embodiments, the calculating of the magnitude of the radial load may be based in part on a position of the support members in the axial direction.

(13) In some embodiments, the radial load may vary over time and the measuring of the at least one of the proportional axial forces applied to the support members and the calculating of the magnitude of the radial load may be repeated periodically over time, or on demand. In this manner, embodiments may provide active, real-time load measurement functionality, or may provide radial load measurement on demand in response by a query of a user or operator, for example.

(14) Although aspects of some of the embodiments disclosed herein are described in the context of measuring a wire load applied to a rotating drum of a winch or a hoist, it is appreciated that aspects may be utilized in a variety of applications to measure a radial load applied to various types of rotating members.

(15) Furthermore, the embodiments disclosed herein may enable accurate measurement of a radial load, regardless of the direction of the radial load and regardless of axial location of the applied radial load. Additionally, the embodiments disclosed herein are configured to determine the radial load without regard to the torque acting on the rotating member.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 shows a perspective view of an exemplary winch, which may include a radial load measuring apparatus according to embodiments described herein.

(2) FIG. 2 shows a cross-sectional schematic view of a winch including a radial load measuring apparatus therein according to an embodiment.

(3) FIG. 3 shows a cross-sectional schematic view of a winch including a radial load measuring apparatus therein according to another embodiment.

(4) FIG. 4 shows a flow chart of a method of measuring a radial load applied to a rotational member.

DETAILED DESCRIPTION

(5) In the following description, certain specific details are set forth in order to provide a thorough understanding of various disclosed embodiments. However, one skilled in the relevant art will recognize that embodiments may be practiced without one or more of these specific details. In other instances, well-known structures and techniques associated with load measuring apparatuses and winches or hoists may not be shown or described in detail to avoid unnecessarily obscuring descriptions of the embodiments.

(6) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

(7) FIG. 1 shows a perspective view of a winch **100**, in which a radial load measuring apparatus may be arranged. It should be understood that radial load measuring apparatuses as disclosed herein are not limited to winches and/or hoists, and may be utilized in a variety of devices with rotational members that experience radial loads. However, the winch **100** is provided as an example

embodiment that may benefit from including a radial load measuring apparatus as described herein.

(8) The winch **100** includes a shaft or drive member **110** defining an axis of rotation **112**. The winch **100** includes a rotating member **114** (e.g., drum) that is rotatably supported on the drive member **110** and is configured to rotate about the axis of rotation **112** by a drive motor (not shown). In use, a cable, rope, wire, or chain (collectively referred to as wire **116** herein) having a first end and a second end opposite the first end may be fastened at the first end thereof to the rotating member **114**. The wire **116** may be wound up about the rotating member **114**, and the second end of the wire **116** may be coupled to a load to be pulled and/or lifted by the winch **100** under the power of the drive motor. The winch **100** can be operated to drive rotation of the rotating member **114** about the drive member **110** with respect to the piece of machinery to which the winch **100** is mounted, such as to wind up the wire **116** onto the rotating member **114** to pull the load toward the winch **100**. When the winch **100** is operated, the winch **100** may be subjected to a radial load, such as in FIG. 1 in which the wire **116** is subjected to a radial load as represented by the arrow labeled L. When the winch **100** is subjected to a radial load L, the radial load is likewise applied to the rotating member **114**.

(9) FIG. 2 shows a cross-sectional schematic view of a winch **100** including an example embodiment of a radial load measuring apparatus **117** when a radial load L1 is applied thereto. As shown in FIG. 2, the radial load measuring apparatus **117** may include at least one support member **118** and at least one force sensor **119**. In some embodiments, two or more support members **118** may be included. In some embodiments one or more of the at least one support member **118** may interface with the rotating member **114** or intervening structure fixedly coupled thereto. It should be understood that in an advantageous embodiment, two support members **118** may be opposite each other in the axial direction D1, such that the rotating member **114** is balanced on two or more support members **118**. In some embodiments, two support members **118** may each interface with one of opposing ends **120**, **121** of the rotating member **114** and the rotating member **114** may be supported in a symmetric or generally symmetric manner on the drive member **110**. In other instances, the rotating member **114** may be supported symmetrically on the drive member **110**. Notably, the drive member **110** is illustrated schematically and details thereof, including structures to drive the rotating member **114** about the axis of rotation **112**, are not shown to avoid unnecessarily obscuring descriptions of the embodiments. Further, although the rotating member **114** is schematically illustrated as mounting to a common elongated drive member **110**, it is appreciated that the rotating member **114** may be supported, either on an externally facing structure or surface or an internally facing structure or surface, by a variety of colinear circular members, such as, for example, a housing bore at one end of the rotating member **114** and outer circumferential surface of a brake housing at the other end of the rotating member **114**.

(10) Each support member **118** may include a contact interface **122**. Each support member **118** may be coupled to the rotating member **114** at the contact interface **122**. In some embodiments, a load transfer member **124** may be arranged between the contact interface **122** and the rotating member **114**. The load transfer member **124** may transfer a load applied to the rotating member **114** to the contact interface **122** and in turn to the support member **118**. The contact interface **122** may be oriented at an oblique angle θ relative to a reference plane **130** that includes the axis of rotation **112**. The oblique angle θ may be, for example, between about 20 degrees and 50 degrees. In embodiments including a plurality of support members **118**, each support member **118** may be configured in some instances to have equal oblique angles θ with respect to each other. Each of the plurality of support members **118** may be configured such that the oblique angles θ oppose one another relative to the reference plane **130**.

(11) When a radial load, such as radial load L1 in FIG. 2, is applied to the rotational member **114** via the wire **116**, the resultant forces L2 arise where the rotational member **114** contacts each of the at least one support member **118**. In some embodiments, such as the embodiment shown in FIG. 2, the resultant forces L2 may arise at opposing ends of the rotational member **114** at the load transfer

members **124** and then are translated from the load transfer members **124** to the support members **118** at the contact interfaces **122**. When the radial load **L1** is applied to the support members **118** in this manner, the contact interfaces **122** are configured to translate the load into proportional axial forces **L3**, due to the oblique angles θ of the contact interfaces **122** of the support members **118** and the application of the resultant forces **L2** acting on the contact interfaces **122**. The proportional axial forces **L3** may be perpendicular or generally perpendicular to the direction of the radial load **L1**. The proportional axial forces **L1** may be parallel with the drive member **110** and axis of rotation **112**. The direction in which the proportional axial forces **L3** are directed is dependent on the configuration of the respective support member **118**. For example, in the embodiment shown in FIG. 2, the support members **118** are configured such that proportional axial forces **L3** are applied away from a center portion **142** of the rotating member **114**. Although the embodiment shown in FIG. 2 shows the support members **118** being configured to apply the proportional axial forces **L3** away from the center portion **142** of the rotating member **114**, the support members **118** may be configured such that the proportional axial forces **L3** are both applied toward the center portion **142** of the rotational member **114**. It should be understood that the radial load **L1** may be equally or substantially equally applied to each of the at least one support member **118** in the case of the illustrated embodiment wherein the support members **118** support the rotational member **114** in a symmetric or generally symmetric manner (i.e., at opposing axial positions). In the case of an asymmetrically supported rotation member **114**, the radial load **L1** may not be equally or substantially equally applied to each support member **118**.

(12) Each support member **118** may comprise at least a portion of a tapered roller bearing assembly **146**. In some embodiments, for example, each support member **118** may be a cone **150** of the tapered roller bearing **146**. The cone **150** may be arranged at least partially within a cup **154** of the tapered roller bearing **146**. In such embodiments, the contact interface **122** may include a face where the cone **150** and the cup **154** interface within the tapered roller bearing assembly **146**. Additionally, the contact interface **122** may include bearing rollers **155** or the like to reduce friction between the cup **154** and the cone **150**. The bearing rollers **155** may be arranged within a cage. When the radial load **L1** is applied to the winch **100** in such embodiments, the radial load **L1** may be transferred to the cups **154** in a radial direction aligned with the radial load **L1** as resultant loads **L2**. The resultant loads **L2** may then be translated to the cones **150** of the tapered roller bearing **146** via the contact interface **122** to become the proportional axial forces **L3**, at least one of which is then applied to the at least one force sensor **119** in the axial direction **D1**.

(13) The at least one force sensor **119** may be configured to measure the proportional axial force **L3** applied by the at least one support member **118**. The at least one force sensor **119** may be coupled to a stationary member **157** such as a backing plate or other rigid structure. The at least one force sensor **119** may be arranged adjacent to the respective support member **118**, such that the support member **118** applies the proportional axial force **L3** to the at least one force sensor **119**. For example, the at least one force sensor **119** may be positioned adjacent to a side **158** of the cup **150** of the tapered roller bearing **146** towards which the proportional axial force **L3** is directed. In some embodiments, the tapered roller bearing **146** may be supported axially by the at least one force sensor **119**, such that the at least one force sensor **119** measures the pressure exerted by the cup **150** of the tapered roller bearing **146** in the axial direction **D1**. In embodiments which include a single force sensor **119**, the support member **118** not associated with the force sensor **119** may be axially supported by a stationary member **157**. In some embodiments, the at least one force sensor **119** may extend around an entire circumference of the drive member **110** or a portion of the rotating member **114**. In some embodiments, the at least one force sensor **119** may only extend around a portion of the drive member **110** or a portion of the rotating member **114**. In some embodiments, the at least one force sensor **119** and/or the stationary members **157** may be adjacent to the drive member **110**. In some embodiments, a gap may be present between the at least one force sensor **119** and/or the stationary member **157** and the drive member **110**.

- (14) In some embodiments, the at least one force sensor **119** may be a capacitive load cell that is configured to interact with the support member **118**. In some embodiments, the at least one force sensor **119** may be a strain gauge load cell that is configured to interact with the support member **118**. Other types of force sensors are also contemplated and the embodiments described herein are not limited to any particular force sensor arrangement.
- (15) The at least one force sensor **119** may be configured to transmit detected force information to a controller. Such a transmission may occur via a wired connection or a wireless connection, including, but not limited to, Bluetooth, Wi-Fi, radio transmission. The controller may include at least one memory for storing instructions and at least one processor for executing the instructions stored in the memory to calculate the radial load **L1** based on the force data detected by the at least one force sensor **119**.
- (16) The proportional axial force **L3** as detected by the at least one force sensor **119** may be used to determine a magnitude of the radial load **L1**. Such a determination may be made by the controller. The relationship between the magnitude of the radial load **L1** and the proportional axial force **L3** may be dependent on a magnitude of the oblique angle θ of the support members **118**. In some embodiments, the proportional axial force **L3** may be directly proportional to the magnitude of the radial load **L1** applied to the rotating member **114**.
- (17) In some embodiments, the cone **150** of each tapered roller bearing **146** may be configured to translate away from the cup **154** in the axial direction **D1** when the radial load **L1** is applied to the cup **154**. The translation of the cone **150** may occur due to the cup **154** acting as a wedge when the radial load **L1** is applied thereto, such that the cup **154** translates in a radial direction aligned with the radial load **L1** and drives the cone **150** in the axial direction **D1** due to the cone **150** being squeezed between the cup **154** and the drive member **110** and/or the rotating member **114**. The tapered roller bearings **146** may include bearing rollers **155** which are configured to allow the cones **150** to translate axially relative to the cups **154**.
- (18) In such embodiments, the at least one force sensor **119** may be configured to measure the proportional axial forces **L3** via displacement of the cones **150** relative to the cups **154**. For example, the at least one force sensor **119** may be a pneumatic or hydraulic piston sensor that is configured to interact with the cone **150** of the tapered roller bearing **146**. As the cone **150** of the tapered support bearing **146** translates in the axial direction **D1**, the cone **150** interacts with the at least one force sensor **119** by displacing the at least one force sensor **119** in the axial direction **D1**. The amount of displacement of the cone **150** may be indicative of the proportional axial force **L3**. The displacement as detected by the at least one force sensor **119** may be used to determine a magnitude of the radial load **L1**. In some embodiments, the displacement of the at least one force sensor **119** may be directly proportional to the magnitude of the radial load **L1** applied to the winch **100**.
- (19) FIG. 3 shows another example embodiment of a cross-sectional schematic view of a winch **200** including an example embodiment of a radial load measuring apparatus **204** when a radial load **L4** is applied thereto. The radial load measuring apparatus **204** is configured similarly to the radial load measuring apparatus **117** of FIG. 2, except as described below. In the embodiment shown in FIG. 3, two support members **208** each interface with one of opposing ends **212**, **216** of a rotating member **220** of the winch **200**. In the embodiment shown in FIG. 3, a drive member **224** configured to rotate about a rotational axis **226** extends into one end **216** of the rotating member **220**, and is coupled to the rotating member **220**, such as by an intervening gear set **227** or other power transmission component, to selectively drive the rotating member **220** about the rotational axis **226** in operation. At one end **212** of the rotating member **220**, the rotating member **220** is supported by a housing bore **228** of a housing **229** via one of the support members **208a**. At the other end **216** of the rotating member **220**, the rotating member **220** is supported by an outer circumferential surface **236** of another component, such as a brake housing **240**, of a brake assembly coupled to or integrated with a drive system of the drive member **224**. Again, components

of the winch **200** (e.g., gear set **227**, brake housing **240**) are illustrated schematically and details thereof are not shown to avoid unnecessarily obscuring descriptions of the embodiments.

(20) With continued reference to FIG. 3, each support member **208** includes a contact interface **248**. In the embodiment shown in FIG. 3, a respective load transfer member **252** is arranged between the contact interface **248** and the rotating member **220** or an intervening structure coupled to the rotating member **220**. The load transfer members **252** transfer a load applied to the rotating member **220** (e.g., a wire load) to the contact interfaces **248** and in turn to the respective support member **208**.

(21) In the embodiment shown in FIG. 3, the winch **200** includes a contact interface **248a** of the support member **208a** which is positioned on a radially-inward side **256** of the respective support member **208a**, and a contact interface **248b** which is positioned on a radially-outward side **260** of the respective support member **208b**. In both cases, the respective support member **208a**, **208b** includes a load transfer member **252** configured to transfer the load **L4** applied to the rotating member **220** to the contact interfaces **248**. The respective support member **218a**, **218b** is configured to receive a resultant force **L5** from the rotating member **220** due to the radial load **L4** in either configuration.

(22) When a radial load, such as radial load **L4** in FIG. 3, is applied to the rotational member **220**, the resultant forces **L5** arise where the rotational member **220** contacts each of the at least one support member **208** via the load transfer members **252**. The resultant forces **L5** may arise at the load transfer members **252** and then are translated from the load transfer members **252** to the support members **208** at the contact interfaces **248**. When the radial load **L4** is applied to the support members **208** in this manner, the contact interfaces **248** are configured to translate the load into proportional axial forces **L6**, due to oblique angles θ of the contact interfaces **248** of the support members **208** and the application of the resultant forces **L5** acting on the contact interfaces **248**. The proportional axial forces **L6** may be perpendicular or generally perpendicular to the direction of the radial load **L1**. In the embodiment shown in FIG. 3, the support members **208** are configured such that proportional axial forces **L6** are applied away from a center portion **262** of the rotating member **220**. Each support member **208** may comprise at least a portion of a tapered roller bearing assembly **264** in a similar fashion to the embodiment shown and described with respect to FIG. 2. Furthermore, at least one force sensor **268** may be configured to measure the proportional axial forces **L6**, similarly to the at least one force sensor **119** shown and described with respect to FIG. 2.

(23) FIG. 4 shows a method **300** of measuring a radial load applied to a rotational member having a radial load measuring apparatus, such as the radial load measuring apparatus described above with reference to FIG. 2. The method **300** begins at step **304** when a radial load **L1** is applied to a winch or hoist **100** including a radial load measuring apparatus **117**, as described herein. Such a radial load **L1** may be applied via a wire **116** extending from the winch, hoist, or other device **100** utilizing the radial load measuring apparatus **117** when the winch, hoist or other device is used to lift or pull an object, for example. The radial load **L1** causes the resultant forces **L2** to arise at the load transfer members **124**. The resultant forces **L2** are then translated to the support members **118** of the radial load measuring apparatus **117**, where the contact interfaces **122** translate the resultant forces **L2** into proportional axial forces **L3**.

(24) During application of a radial load **L1**, at step **308**, at least one of the proportional axial forces **L3** is measured. At least one of the proportional axial forces **L3** is applied to at least one force sensor **119**. The at least one force sensor **119** may obtain then transmit this information to a controller of the radial load measuring apparatus **117**.

(25) At step **312**, a magnitude of the radial load **L1**, based at least in part on the measurement of the proportional axial force **L3**, is calculated. Such a calculation may be dependent on a magnitude of an oblique angle θ of the contact interface **122** of the support members **118** and other factors, such as the axial positions of the support members **118**. The calculating of the radial load **L1** may be

performed by the controller of the radial load measuring apparatus **117**. A calculation for determining the radial load **L1** may be stored in a memory of the controller and the radial load **L1** determined by the controller may be stored in the memory of the controller. In some instances, the radial load **L1** may be displayed to a user or operator by a display screen or the like.

(26) In some embodiments, the controller may be configured to output a message in response to the radial load **L1** determined. For example, if the radial load **L1** exceeds a predetermined threshold, which may be less than or equivalent to a maximum load rating for the winch **100**, the controller may output a warning message for an excessive radial load **L1**. In another example, if the radial load **L1** exceeds a predetermined maximum load, the controller may carry out certain actions, such as, for example, causing the motor to cease providing power to the winch **100**. In another example, depending on the determined the radial load **L1**, the winch **100** may shift into a different ratio or motor displacement to change the load capacity of the winch **100**. In a further example, if the radial load **L1** exceeds a predetermined threshold, certain functions of the winch **100** may be limited.

(27) Over time, the radial load **L1** may vary. Therefore, the steps **308** and **312** may be repeated continuously or periodically over time. Each determined radial load **L1** or a history of the radial load **L1** may be stored in the memory of the controller.

(28) The method **300** may end at step **316** when a radial load **L1** ceases to be applied to the rotational member **114**.

(29) The operation of the controller described above in method **300** may be executed by a processor of the controller.

(30) The radial load measuring apparatus and associated methods described herein may be adapted to a variety of different pieces of equipment or machinery, including those with winches and/or hoists. Such a radial load measuring apparatus may provide an improved testing solution when the load-handling capabilities of a winch, hoist, or other rotatable device is considered. Such a radial load measuring apparatus may provide enhanced functionality for a user or operator of the host equipment or machinery to operate the equipment or machinery in a reliable and safe manner.

(31) The devices and systems of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features described above may be used independently of one another, or may be combined in various ways. All possible combinations and subcombinations are intended to fall within the scope of this disclosure. Various modifications to the implementations described in this disclosure may be readily apparent to those of ordinary skill in the art, and the generic principles defined herein may be applied to other implementations. Thus, the claims are not intended to be limited to the implementations shown herein, but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features disclosed herein.

(32) Certain features that may be described in this specification in the context of separate implementations also may be implemented in combination in a single implementation. Conversely, various features that may be described in the context of a single implementation also may be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination. No single feature or group of features is necessary or indispensable to each and every embodiment.

(33) Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments. The terms “comprising,” “including,” “having,” and the

like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise.

(34) In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled.

Claims

1. A radial load measuring apparatus comprising: a rotational member rotatably supported to rotate about an axis of rotation and configured to be subjected to a radial load; at least one support member interfacing with the rotational member and having a contact interface that is oriented at an oblique angle relative to a reference plane comprising the axis of rotation such that the radial load applied to the rotational member is translated into a proportional axial force that is applied to the at least one support member in an axial direction; and a force sensor configured to measure the proportional axial force applied to the at least one support member from which to determine a magnitude of the radial load.
2. The radial load measuring apparatus of claim 1, wherein the at least one support member comprises at least a portion of a taper roller bearing assembly.
3. The radial load measuring apparatus of claim 1, wherein the at least one support member comprises a cone of a taper roller bearing assembly, and wherein the rotational member is fitted with a cup of the taper roller bearing assembly and wherein, during operation, the cone of the taper roller bearing assembly is able to translate away from the cup of the taper roller bearing assembly.
4. The radial load measuring apparatus of claim 1, wherein the force sensor is configured to measure the proportional axial force applied to the at least one support member via displacement of the cone of the taper roller bearing assembly relative to the cup of the taper roller bearing assembly.
5. The radial load measuring apparatus of claim 1, wherein the force sensor is a capacitive load cell that interacts with the at least one support member.
6. The radial load measuring apparatus of claim 1, wherein the force sensor is a pneumatic or hydraulic piston sensor that interacts with the at least one support member.
7. The radial load measuring apparatus of claim 1, wherein the force sensor is a strain gauge load cell that interacts with the at least one support member.
8. The radial load measuring apparatus of claim 1, wherein the proportional axial force applied to the at least one support member that is measured by the force sensor is directly proportional to a magnitude of the radial load applied to the rotational member.
9. The radial load measuring apparatus of claim 1, wherein the rotational member comprises a drum of a winch or hoist, and wherein the radial load applied to the drum is a wire load.
10. The radial load measuring apparatus of claim 1, wherein the radial load measuring apparatus comprises a pair of support members, wherein each support member interfaces with a respective one of the opposing ends of the rotational member.
11. The radial load measuring apparatus of claim 1, wherein the radial load measuring apparatus comprises a pair of support members, wherein the oblique angle of the contact interface of each support member is uniform.
12. An apparatus for measuring a radial wire load applied to a winch or hoist comprising: a drum rotatably supported to rotate about an axis of rotation and configured to be subjected to a radial wire load; a support member interfacing with one of opposing ends of the drum and having a

contact interface that is oriented at an oblique angle relative to a reference plane comprising the axis of rotation such that the radial wire load applied to the drum is translated into a proportional axial force applied to the support member; and a force sensor configured to measure the proportional axial force applied to the support member from which to determine a magnitude of the radial wire load.

13. A method of measuring a radial load applied to a rotational member rotatably supported to rotate about an axis of rotation, the rotational member being supported by at least one support member, the at least one support member interfacing with the rotational member and the at least one support member having a contact interface that is oriented at an oblique angle relative to a reference plane comprising the axis of rotation such that the radial load applied to the rotational member is translated into a proportional axial force that is applied to of the at least one support member in an axial direction, the method comprising: during application of the radial load, measuring the proportional axial force applied to one or more of the at least one the support member; and calculating a magnitude of the radial load based at least in part on the measurement of the proportional axial force.

14. The method of claim 13, wherein the calculating of the magnitude of the radial load is based in part on the oblique angle of the contact interface of the at least one support member.

15. The method of claim 13, wherein the radial load varies over time and wherein the measuring of the proportional axial force applied to the at least one support member and the calculating of the magnitude of the radial load is repeated periodically over time.
